

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

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COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">  
  <input type="text" name="name">  
  <input type="email" name="email">  
  <button>Submit</button>  
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure. <header> – Displays the page heading.

- <main> – Contains the main content.
- <section> – Groups related content.
- <form> labels (<label>) – Identifies input fields.

<footer> (optional) – Displays footer information

2. Modify the structure to make it **standards-compliant**.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Online Symposium Registration</title>
```

```
</head>
```

```
<body>
```

```
    <header>
```

```
        <h1>Online Symposium Registration</h1>
```

```
    </header>
```

```
    <main>
```

```
        <section>
```

```
            <form action="/register" method="POST">
```

```
                <label for="name">Student Name:</label>
```

```
<input type="text" id="name" name="name" required>
```

```
<br><br>
```

```
<label for="email">Email:</label>
```

```
<input type="email" id="email" name="email" required>
```

```
<br><br>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
</section>
```

```
</main>
```

```
<footer>
```

```
<p>&copy; 2026 College Symposium</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

3. Interpret the **HTTP request generated** during form submission. POST /register HTTP/1.1
Host: www.college.edu

Content-Type: application/x-www-form-urlencoded
name=John+Doe&email=john@example.com

Explanation:

- **POST** – Sends data securely in the request body.
- **/register** – Server endpoint handling registration.
- **Host** – Specifies the website.
- **Content-Type** – Indicates form data format.
- **Body** – Contains the student's name and email.

4. Analyze the **HTTP response cycle** from server to browser.

1. Student fills out the form.
2. Browser sends a **POST** request to `/register`.
3. Server receives and validates the data.
4. Server stores the registration in the database.
5. Server generates an HTTP response.
6. Browser receives the response and displays the result.

Example response:

HTTP/1.1 200 OK
Content-Type: text/html

```
<html>
<body>
<h2>Registration Successful!</h2>
</body>
</html>
```

Possible status codes:

- **200 OK** – Registration successful.
- **400 Bad Request** – Invalid input.
- **404 Not Found** – Endpoint not found.
- **500 Internal Server Error** – Server error.

5. Suggest improvements for **usability and accessibility**.

- Use `<label>` elements for all input fields.
- Add the `required` attribute for mandatory fields.
- Use appropriate input types (email, text).
- Include placeholders (optional) to guide users.
- Add ARIA attributes where necessary for assistive technologies.
- Ensure keyboard navigation is supported.

- Use clear error messages for invalid inputs.
- Make the page responsive using the viewport meta tag.
- Use semantic HTML elements (header, main, section, footer) for better structure.

Result: These improvements make the symposium registration portal more standards-compliant, user-friendly, accessible, and easier to maintain.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
<head>
<title>Company</title>
</head>
<body>
<div>
<h1>Welcome
<p>About us
</div>
</body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.

The given HTML contains several errors:

- Missing `</h1>` closing tag.
- Missing `</p>` closing tag.
- Missing `<!DOCTYPE html>` declaration.
- Missing `<meta charset="UTF-8">`.
- Missing `<meta name="viewport">`.
- Uses a generic `<div>` instead of semantic elements like `<header>` and `<section>`.

2. Analyze how different browsers may interpret this structure.

Since the HTML is not properly closed, browsers try to correct it automatically.

- **Google Chrome:** Automatically closes missing tags and displays the page correctly in most cases.
- **Mozilla Firefox:** Repairs the HTML but may render spacing or layout differently.
- **Microsoft Edge:** Behaves similarly to Chrome but minor differences may occur.
- **Safari:** Attempts to fix the structure but rendering can vary depending on the browser engine.

Because browsers have different HTML parsers, improperly written HTML can lead to inconsistent layouts.

3. Identify **missing semantic elements** (e.g., header, section). The page should include the following HTML5 semantic elements:

- <header> – Contains the page heading.
- <main> – Main content area.
- <section> – Groups related content such as "About Us".
- <footer> (optional) – Footer information.

4. Provide a **corrected W3C-compliant HTML structure**.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Company</title>
</head>
<body>

  <header>
    <h1>Welcome</h1>
  </header>

  <main>
    <section>
      <p>About us</p>
    </section>
  </main>

  <footer>
    <p>&copy; 2026 Company</p>
  </footer>

</body>
</html>
```

5. Suggest techniques to ensure **cross-browser compatibility**.

- Write **valid W3C-compliant HTML**.
- Use proper semantic HTML5 elements.
- Close all HTML tags correctly.
- Include <!DOCTYPE html> to enable standards mode.
- Test the website on multiple browsers (Chrome, Firefox, Edge, Safari).
- Use CSS resets or Normalize.css to reduce browser styling differences.
- Use responsive design with the viewport meta tag.

- Validate HTML and CSS using W3C validators.
- Avoid browser-specific features unless necessary.
- Use feature detection and provide fallbacks for unsupported features.

Result

Following HTML5 standards, using semantic elements, validating code, and testing across multiple browsers ensures consistent layout and behavior on all major browsers.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.

The form has the following problems:

- The button uses `type="button"` instead of `type="submit"`, so clicking it does **not** submit the form.
- The **GET** method is used for a login form, which is **not secure** because credentials are sent in the URL.
- Passwords should never be transmitted using the GET method.

2. Analyze why the **HTTP request is not triggered**.

The HTTP request is not sent because:

- `<button type="button">` only displays a clickable button.
- It does **not** submit the form automatically.
- Since no JavaScript is attached to the button, clicking it performs no action.

3. Interpret the role of **GET method in this scenario**.

The **GET** method:

- Sends form data in the URL as query parameters.
- Is mainly used for retrieving data from the server.

- Is **not suitable for login forms** because usernames and passwords become visible in:
 - Browser address bar
 - Browser history
 - Server logs
 - Bookmarks

Example of a GET request:

```
GET /submit?username=john&password=12345 HTTP/1.1
```

This exposes sensitive information, making GET inappropriate for authentication.

4. Propose a **corrected implementation**.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login Form</title>
</head>
<body>

  <form action="/submit" method="POST">

    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required>

    <br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required>

    <br><br>

    <button type="submit">Login</button>

  </form>

</body>
</html>
```

5. Suggest improvements for **secure data transmission**.

Use the **POST** method instead of GET.

Use `type="submit"` for the login button.

Enable **HTTPS** to encrypt data during transmission.

Validate inputs on both the client and server.

Store passwords securely using hashing (never store plain text).
 Implement CSRF protection for forms.
 Use strong authentication practices (e.g., secure sessions, optional multi-factor authentication).
 Display clear error messages without revealing sensitive information.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis

Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and	Mostly accurate with	Partially correct with	Incorrect answers

		covers all required points	minor omissions	missing points	with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand